| | | | |
|---|---|---|---|
| **Teacher:** | Ray Joshua Balingkit | **Subject:** | Computing |
| **Week:** | 4 (Term 1) | **Year:** | 7 |
| **Link:** | | **Cover:** | |

## LESSON OVERVIEW

### What should students know/do by the end of this week's lessons?

Students will learn to distinguish numeric values from digit characters and encode simple text using a supplied character table. They will also learn to describe how bitmap images and sampled sound can be represented as digital data. Through teacher demonstrations, discussion and practical activities, students will practise these skills and explain their ideas.

### End-of-Week Assessment/Check: How will you know they got it?

Use teacher observation, pupil explanations and the completed practical or paper activity. Check accurate codes and an explanation of why a digit character differs from its numeric value; do not teach ASCII as 256 characters. Check the pixel order and colour key, then ask what each sound sample measures at a particular time. Note pupils who need further guided practice.

## LESSON SLOTS - MONDAY AND TUESDAY

| Lesson | Date | Learning Objective | The Lesson Flow | Differentiation |
|---|---|---|---|---|
| 1 | 14 September 2026 | By the end of this lesson, students will be able to distinguish numeric values from digit characters and encode simple text using a supplied character table. | **H:** Compare the number 12 used in a calculation with the characters in a fictional code 12.<br><br>**ID:** Explain character encoding. Clarify standard ASCII has 128 codes; this class displays them in eight-bit form with a leading zero. Unicode supports more scripts.<br><br>**WD:** Compare numeric 5 as 00000101 with character '5' as 00110101, then encode 'a' as 01100001.<br><br>**YD:** Pupils encode and decode a short non-personal word, spaces and digit characters using the provided table.<br><br>**C:** Check accurate codes and an explanation of why a digit character differs from its numeric value; do not teach ASCII as 256 characters. | **Support:** Provide place-value grids, carry rows, a character-code table and labelled diagrams. Model one step at a time and allow spoken reasoning before writing.<br><br>**Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation. |
| 2 | 15 | By the end of this lesson, | **H:** Zoom into a small | **Support:** Provide place-value |

| | | | | |
|---|---|---|---|---|
| | September 2026 | students will be able to describe how bitmap images and sampled sound can be represented as digital data. | bitmap and display a simple sound-wave diagram.<br><br>**ID:** Introduce pixels, a colour key and regular measurements of sound; distinguish an image grid from an audio sample sequence.<br><br>**WD:** Encode a four-by-four two-colour image, then record five readings at equal intervals on a prepared wave graph.<br><br>**YD:** Pupils decode a small bitmap and annotate a sound-sampling diagram, explaining what the stored numbers represent.<br><br>**C:** Check the pixel order and colour key, then ask what each sound sample measures at a particular time. | grids, carry rows, a character-code table and labelled diagrams. Model one step at a time and allow spoken reasoning before writing.<br><br>**Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation. |
| **3** | | | | Support:<br><br>Extension: |
| **4** | | | | Support:<br><br>Extension: |

## MATERIALS NEEDED FOR THE WEEK

- Oxford International Lower Secondary Computing Student Book 7 printed pages 22-29
- Teacher computer and display
- Classroom computers or laptops and the class save folder
- Binary place-value grids, carry rows, character-code tables and pixel or sampling sheets
- Whiteboard and markers
- Student notebooks, pencils and task sheets
- Teacher reference: Unicode Consortium glossary, https://unicode.org/glossary/



## END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?